3 Bayesian Reasoning

The last chapter outlined a common approach of developing a supervised learning algorithm using a hypothesis space, an error metric, and optimization. This is called a discriminative approach and is one we will explore in later chapters. However, before addressing it, we will first cover in this chapter the **generative approach** in which we model the joint distribution between features ($\vec{x}$) and label (y).

Further information
To avoid concept confusion, it is important to understand the structure of this chapter before diving in.

We will first discuss an *ideal* but *unrealizable* classifier, the Bayes-optimal classifier. We will then consider approximating this method. Through a series of two approximations, we will arrive at the achievable **naive Bayes classifier**.

This chapter is concerned with classification: The output, y, takes on one of a few distinct categories. While these categories could be anything (for example two categories: will click on the advertisement, will not click on the advertisement), we will denote each of the categories with a different integer. We let C be the number of categories, and thus $y \in \{1, 2, \ldots, C\}$.

Bayes-Optimal Classifier

Our assumption is that there is a single distribution generating pairs of inputs and outputs ($\vec{x}$ and y). We will denote this joint distribution as $\Pr(\vec{x}, y)$. While this distribution is unknown, we can consider what to do *if* it were known.

Our goal is to generate a function that predicts y from $\vec{x}$. The correct output may not be completely knowable from the input. Consider trying to estimate the disease of your hippopotamus patient. We might have three features, $\vec{x}$: whether or not the patient has a cough, whether or not the patient has a runny nose, and whether or not the patient has a headache. If a hippopotamus arrives in our office with a cough, no runny nose, and no headache, we do not completely know whether the hippopotamus has the common cold (one possible value for y) or just bad allergies (another possible value).

Therefore, in order to do as best as we can, we can select the value for y that is *most likely*, given the observations we have made, $\vec{x}$. Thus, we want to pick the value of y that makes $\Pr(y \mid \vec{x})$ as large as possible (recall that $\vec{x}$ is known). We would could write this as

$$\begin{aligned}
f_{\text{BayesOpt}}(\vec{x}) &= \operatorname*{arg\,max}_{c\in\{1,2,\ldots,C\}} \boxed{\Pr(y{=}c \mid \vec{x})} && \text{Bayes' rule} \\
&= \operatorname*{arg\,max}_{c\in\{1,2,\ldots,C\}} \boxed{\frac{\Pr(\vec{x} \mid y{=}c)\Pr(y{=}c)}{\Pr(\vec{x})}} && \text{denominator does not depend on } c \\
&= \operatorname*{arg\,max}_{c\in\{1,2,\ldots,C\}} \boxed{\Pr(\vec{x} \mid y{=}c)\Pr(y{=}c)} \,. && (3.1)
\end{aligned}$$

The arg max indicates that the result is the argument that maximizes the value to the right ($\Pr(y{=}c \mid \vec{x})$ in this case). The $c \in \{1, 2, \ldots, C\}$ indicates that it is the variable c that we will be adjusting to find the maximum and that it can only be one of the values in the set $\{1, 2, \ldots, C\}$, the integers from 1 through C. So the first line states that our function f should return the class (c) that corresponds to the maximum value of the probability of y being that class, given our observation vector, $\vec{x}$. We then transform this expression by applying Bayes' rule, and then noting that the denominator of the expression does not depend on c. Therefore the value of c that maximizes the full fraction will be the same as the value of c that maximizes the numerator.

This maximization is simple: We can just try each possible value for the predicted class, c, and remember for which one $\Pr(\vec{x} \mid y = c)\Pr(y = c)$ is largest. The first component is the probability of observing these features, given that the class is (hypothetically) c. The second is the prior probability that the class is c.

> **Further information**
> Yes, $\Pr(\vec{x} \mid y{=}c)\Pr(y{=}c)$ is the same as $\Pr(\vec{x}, y = c)$, but we are going to need this form so we will keep it as the product.

We call this classification rule, f_{BayesOpt}, the **Bayes-optimal classifier**. It selects the prediction that maximizes the probability it is correct (and thereby minimizes the probability it is not the true value for y). It does not depend on any data, but instead requires knowledge of the true distribution $\Pr(\vec{x}, y)$. Thus, it is an ideal and something (at least theoretically) to compare against.

> **Key point**
> The Bayes-optimal classifier is a generally unobtainable function that represents the best possible result.

An Example

To explore this approach, consider trying to find your favorite hippopotamus, Sibonelo. You would like to predict his location based on the weather. There are only three weather types: clear, overcast, and rainy. Sibonelo can only be in one of two locations: at the lake, or in the grassy field.

In general, we do not know the distribution from which the data were sampled. But, for the sake of this example, we will assume it is as below.

$$\begin{array}{c} \Pr(\vec{x},y) \\ \begin{array}{r} \\ \text{clear} \\ \text{overcast} \\ \text{rainy} \end{array} \begin{array}{cc} \text{lake} & \text{grass} \\ \left[\begin{array}{cc} 0.35 & 0.10 \\ 0.13 & 0.14 \\ 0.10 & 0.18 \end{array}\right] \end{array} \end{array} \Rightarrow \begin{array}{c} \Pr(\vec{x}) \\ \begin{array}{r} \text{clear} \\ \text{overcast} \\ \text{rainy} \end{array} \left[\begin{array}{c} 0.45 \\ 0.27 \\ 0.28 \end{array}\right] \end{array} \Rightarrow \begin{array}{c} \Pr(y \mid \vec{x}) \\ \begin{array}{r} \\ \text{clear} \\ \text{overcast} \\ \text{rainy} \end{array} \begin{array}{cc} \text{lake} & \text{grass} \\ \left[\begin{array}{cc} 0.78 & 0.22 \\ 0.48 & 0.52 \\ 0.36 & 0.64 \end{array}\right] \end{array} \end{array}$$

$$\Downarrow$$

$$\Pr(y) \quad \begin{array}{cc} \text{lake} & \text{grass} \\ \left[\begin{array}{cc} 0.58 & 0.42 \end{array}\right] \end{array}$$

$$\Downarrow$$

$$\Pr(\vec{x} \mid y) \quad \begin{array}{r} \\ \text{clear} \\ \text{overcast} \\ \text{rainy} \end{array} \begin{array}{cc} \text{lake} & \text{grass} \\ \left[\begin{array}{cc} 0.60 & 0.24 \\ 0.22 & 0.33 \\ 0.17 & 0.43 \end{array}\right] \end{array} \tag{3.2}$$

The joint distribution is in the upper left. All of the other distributions (the two marginals, and the two conditionals) can be calculated from the joint and are shown just as a reminder of how these different views of the same distribution relate.

The Bayes-optimal classifier would use the table for $\Pr(y \mid \vec{x})$ to make its prediction. If the weather is overcast, $\Pr(\text{grass} \mid \text{overcast}) = 0.52$ which is greater than $\Pr(\text{lake} \mid \text{overcast}) = 0.48$. Therefore, the Bayes-optimal prediction is that Sibonelo would be found in the lake on an overcast day. With similar logic, the Bayes-optimal classifier would report “lake” for clear days and “grass” for rainy days. Note that it is *not* correct to use the table for $\Pr(\vec{x} \mid y)$. While for this dataset, we would still produce the same answer (grass), this is not always true. We want to condition on what we have observed: $\vec{x}$.

Approximate Bayes-Optimal Classifier

We do not know $\Pr(\vec{x}, y)$, so we cannot use this ideal. Yet, even if we cannot calculate the Bayes-optimal classifier, we can approximate it using data. For this example, consider that we have collected the training set shown in Table 3.1.

We can use these data to estimate the true distribution, and then we can use this estimate instead of the true distribution in the formula for the Bayes-optimal classifier, Equation 3.1.

weather	location
overcast	grass
overcast	grass
clear	lake
rainy	lake
rainy	grass
rainy	grass
overcast	lake
clear	lake
clear	lake
clear	lake

Table 3.1: Sibonelo-spotting data

Empirical Distribution

The **empirical distribution** is the distribution calculated by counting the frequency at which the events happened in the (training) data. As the amount of training data grows, the empirical distribution will approach the true distribution. For the data in Table 3.1, the empirical distribution is

$$
\begin{array}{r|cc}
\widehat{\Pr}(\vec{x}, y) & \text{lake} & \text{grass} \\
\text{clear} & 0.40 & 0.00 \\
\text{overcast} & 0.10 & 0.20 \\
\text{rainy} & 0.10 & 0.20
\end{array}
\; . \tag{3.3}
$$

There are ten examples (conveniently) in our data, so these probabilities have simple decimal values.

From this we can calculate all of the marginals and conditionals, just as we could for the true distribution (see Equation 3.2). In particular, the conditional we are interested in is

$$
\begin{array}{r|cc}
\widehat{\Pr}(y \mid \vec{x}) & \text{lake} & \text{grass} \\
\text{clear} & 1.00 & 0.00 \\
\text{overcast} & 0.33 & 0.67 \\
\text{rainy} & 0.33 & 0.67
\end{array}
\; . \tag{3.4}
$$

We can use this empirical distribution instead of the true (generally unknown) distribution. Our final classifier would report "lake" if the weather is clear ($1.00 > 0.00$). And, it would report "grass" if the weather is overcast ($0.67 > 0.33$) or rainy ($0.67 > 0.33$)

Further information
If the empirical distribution were even, for example if it were 0.5 for both $\widehat{\Pr}(y = \text{lake} \mid \vec{x})$ and $\widehat{\Pr}(y = \text{grass} \mid \vec{x})$, then we could break the tie in any way we desired.

Naive Bayes Classifier

The method above follows the generative approach for supervised learning: We estimated the joint distribution of $\vec{x}$ and y and then used this estimated joint distribution as if it were the real distribution to make our decision or classification. For a simple dataset like the one above, our estimate of the joint distribution can just be the empirical distribution. But, if the problem is more complex, we run into trouble using the empirical distribution.

Consider if we have collected more features that might be helpful in predicting where to find Sibonelo, resulting in the dataset with three features shown in Table 3.2.

weather	time	moon	location
overcast	day	full	grass
overcast	night	full	grass
clear	day	new	lake
rainy	night	full	lake
rainy	night	full	grass
rainy	day	new	grass
overcast	day	new	lake
clear	night	full	lake
clear	night	full	lake
clear	night	new	lake

Table 3.2: Full Sibonelo-spotting data

Constructing the empirical joint distribution and resulting conditional distribution, we get

$$
\begin{array}{r}
\widehat{\Pr}(\vec{x},y) \\
\text{clear, day, full} \\
\text{clear, day, new} \\
\text{clear, night, full} \\
\text{clear, night, new} \\
\text{overcast, day, full} \\
\text{overcast, day, new} \\
\text{overcast, night, full} \\
\text{overcast, night, new} \\
\text{rainy, day, full} \\
\text{rainy, day, new} \\
\text{rainy, night, full} \\
\text{rainy, night, new}
\end{array}
\begin{array}{c}
\begin{array}{cc} \text{lake} & \text{grass} \end{array} \\
\begin{bmatrix}
0.00 & 0.00 \\
0.10 & 0.00 \\
0.20 & 0.00 \\
0.10 & 0.00 \\
0.00 & 0.10 \\
0.10 & 0.00 \\
0.00 & 0.10 \\
0.00 & 0.00 \\
0.00 & 0.00 \\
0.00 & 0.10 \\
0.10 & 0.10 \\
0.00 & 0.00
\end{bmatrix}
\end{array}
\Rightarrow
\begin{array}{r}
\widehat{\Pr}(y \mid \vec{x}) \\
\text{clear, day, full} \\
\text{clear, day, new} \\
\text{clear, night, full} \\
\text{clear, night, new} \\
\text{overcast, day, full} \\
\text{overcast, day, new} \\
\text{overcast, night, full} \\
\text{overcast, night, new} \\
\text{rainy, day, full} \\
\text{rainy, day, new} \\
\text{rainy, night, full} \\
\text{rainy, night, new}
\end{array}
\begin{array}{c}
\begin{array}{cc} \text{lake} & \text{grass} \end{array} \\
\begin{bmatrix}
? & ? \\
1.00 & 0.00 \\
1.00 & 0.00 \\
1.00 & 0.00 \\
0.00 & 1.00 \\
1.00 & 0.00 \\
0.00 & 1.00 \\
? & ? \\
? & ? \\
0.00 & 1.00 \\
0.50 & 0.50 \\
? & ?
\end{bmatrix}
\end{array}
$$

There are 12 possible values for $\vec{x}$, corresponding to the different combinations of weather, time, and moon phase. Because we have only 10 examples in our training data, some of these combinations have never been observed.[1] This leads to rows in the joint distribution with all zeros. We cannot estimate the conditional distribution for the same row. (We would divide zero by zero.)

[1] In fact, only 8 unique $\vec{x}$ values have been observed. On a rainy night with a full moon, Sibonelo has been observed both at the lake and in the grass. And, Sibonelo has been observed twice on a clear night with a full moon, both times in the lake.

For instance, if we wish to find Sibonelo on a rainy day with a full moon, looking at our training data, we have never observed this combination before and therefore the empirical distribution has nothing to say about whether finding Sibonelo in the lake or the grass is more likely. Further, most of the rows (values for $\vec{x}$) have only a single observation. This means we are estimating where to find Sibonelo from only a single previous example. For instance, on a rainy day with a new moon, we would predict Sibonelo would be in the grass. Yet, every other time it was a new moon, Sibonelo was found in the lake. It might be that a rainy day with a new moon is completely unrelated to other times it is a new moon. But, without more data to support that, it would seem wise to be able to also consider other past situations where $\vec{x}$ was similar, but not exactly the same.

We might consider solving this problem by collecting more data. But, the number of values of $\vec{x}$ grows geometrically with the number of features. If we have n binary features,[2] there are 2^n possible values for $\vec{x}$. This means that for even a moderate number of features, collecting enough data to sufficiently populate the joint empirical distribution is impossible.

[2] A binary feature is one that can take on only two values, like moon phase or time in our example.

Assumptions and Approximations

We might generate any number of ad-hoc methods for circumventing this problem and using "similar" historic examples to aid in classifying the current example. However, we much prefer to specify a few, easily stated assumptions, and derive the rest of the method as a consequence of these assumptions. Then we can, as human beings, more easily evaluate whether the method is suitable to our problem by examining how closely we think our problem matches the assumptions.

For the naive Bayes classifier, the assumption is that *the features are independent, given the label*. Put mathematically, our assumption about the true joint distribution is that

$$\begin{aligned}
\Pr(\vec{x} \mid y) &= \Pr(x_1, x_2, \ldots, x_n \mid y) \\
&= \Pr(x_1 \mid y)\Pr(x_2 \mid y) \cdots \Pr(x_n \mid y) \quad \text{the naive Bayes assumption} \\
&= \prod_{j=1}^{n} \Pr(x_j \mid y) \qquad (3.5)
\end{aligned}$$

This does not say that the features are independent (a different assertion). It says that if we know the value of y (the class label), then each of the features are independent. As an example, this means that given we know Sibonelo is at the lake, the weather and whether it is day or night are independent. Thus, it might be that it is more likely to rain at night (thus the weather and time are dependent). However, if we know Sibonelo is at the lake, the assumption is that the two are independent given this condition.

This might seem unnatural in this scenario, as we would generally assume that it is the weather and time that cause the location of Sibonelo, not the reverse. Dependence or independence of random variables need not conform to causality. But, to make the situation more clear, consider a different example: We are trying to predict whether Sibonelo has a cold based on two observations: whether Sibonelo is sneezing and whether Sibonelo is wiggling his ears.[3] These two features are dependent (because being sick causes both more sneezing and less ear wiggling): If we see a sneezing hippopotamus we expect less ear wiggling. However, it might be reasonable to assume that both symptoms manifest independently, given a cold (or given no cold). That is the naive Bayes assumption.

[3] It is well known that hippopotamuses wiggle their ears less when they feel sick.

Armed with the assumption of Equation 3.5, we can derive the naive Bayes algorithm. We first decompose the joint distribution according to the assumption. We then replace true distribution with the empirical one. Starting from the Bayes-optimal criterion of Equation 3.1, we have the following.

$$\begin{aligned}
&\operatorname*{arg\,max}_{c\in\{1,2,\ldots,C\}} \Pr(\vec{x} \mid y=c)\Pr(y=c) \\
&\qquad= \operatorname*{arg\,max}_{c\in\{1,2,\ldots,C\}} \left(\prod_{j=1}^{n} \Pr(x_j \mid y=c)\right)\Pr(y=c) && \text{the naive Bayes assumption} \\
&\qquad\approx \operatorname*{arg\,max}_{c\in\{1,2,\ldots,C\}} \left(\prod_{j=1}^{n} \widehat{\Pr}(x_j \mid y=c)\right)\widehat{\Pr}(y=c) && \text{using the empirical distribution} \\
&\qquad= f_{\text{NB}}(\vec{x}) && (3.6)
\end{aligned}$$

where the final line indicates that this is the naive Bayes (NB) classifier, denoted f_{NB}.

We will break this computation into two parts: calculating the part after the arg max (which we will call the discriminant) and performing the arg max (finding the value of c that maximizes the discriminant):

$$f_{\text{NB}}(\vec{x}) = \operatorname*{arg\,max}_{c\in\{1,2,\ldots,C\}} f_c(\vec{x}) \qquad f_c \text{ is the discriminant for class } c \tag{3.7}$$

$$f_c(\vec{x}) = \widehat{\Pr}(y=c)\prod_{j=1}^{n}\widehat{\Pr}(x_j \mid y=c) \qquad \text{putting the } \widehat{\Pr}(y=c) \text{ term first} \tag{3.8}$$

While we still are using the empirical distribution, it is not the full empirical distribution over all joint possible values for $\vec{x}$. Instead, we have one small conditional distribution for each feature, just over the possible values of that feature (and the target): $\widehat{\Pr}(x_j \mid y)$. So, if there are n binary features, this means we need only n such distributions, each with a small number of probabilities. Thus, the representation size grows linearly with the number of features, rather than geometrically when using the full joint distribution.

In return, we gave up the ability to model dependencies among the features conditioned on the label. If two features are related, even when conditioned on the label (like the girth and weight of a hippopotamus conditioned on the species), by assuming they are independent, this method will over weigh the significance of the evidence. This is akin to the situation in which you hear that "hippopotamuses produce pink milk"[4] from two different friends. If you assume these are independent sources, you would place greater confidence in the truth of the statement than if you learned that they both heard the rumor from the same source.

[4] which is not true

Example Calculations

We can see from Equation 3.8, that the final classifier, f_{NB}, depends on the marginal conditional empirical distribution[5] of one feature conditioned on the label. It also depends on the marginal distribution over the label. Thus, our learning procedure is to take the the training data and produce these empirical distributions. Armed with these distributions, we no longer need the data and can calculate the prediction for any new situation.

[5] Why all these modifiers?

marginal: all but one feature have been marginalized out

conditional: conditioned on y

empirical: frequencies in the data

For the data of Table 3.2, the needed emprirical distributions are

$$\begin{array}{c|cc} \widehat{\Pr}(y) & \text{lake} & \text{grass} \\ \hline & 0.60 & 0.40 \end{array} \qquad \begin{array}{c|cc} \widehat{\Pr}(x_1 \mid y) & \text{lake} & \text{grass} \\ \hline \text{clear} & 0.67 & 0.00 \\ \text{overcast} & 0.17 & 0.50 \\ \text{rainy} & 0.17 & 0.50 \end{array}$$

y=lake for 6 of the 10 examples

x_1=overcast for 2 of the 4 examples where y=grass

$$\begin{array}{c|cc} \widehat{\Pr}(x_2 \mid y) & \text{lake} & \text{grass} \\ \hline \text{day} & 0.33 & 0.50 \\ \text{night} & 0.67 & 0.50 \end{array} \qquad \begin{array}{c|cc} \widehat{\Pr}(x_3 \mid y) & \text{lake} & \text{grass} \\ \hline \text{full} & 0.50 & 0.75 \\ \text{new} & 0.50 & 0.25 \end{array} \tag{3.9}$$

Further information
The conditional distributions needed for naive Bayes are of the form $\widehat{\Pr}(x_j \mid y)$, *not* $\widehat{\Pr}(y \mid x_j)$; do not confuse the two. Naive Bayes is based on $\widehat{\Pr}(y \mid \vec{x})$ but it uses $\widehat{\Pr}(x_j \mid y)$ to calculate it.

These tables constitute everything our classifier needs to make predictions.

Consider now that it is an overcast day with a full moon (that is, $\vec{x} = \begin{bmatrix}\text{overcast}\\ \text{day}\\ \text{full}\end{bmatrix}$). Where would our naive Bayes classifier predict we will find Sibonelo? We calculate the two (because there are two classes) discriminants: $f_{\text{lake}}(\vec{x})$ and $f_{\text{grass}}(\vec{x})$ as in Equation 3.8. Equation 3.7 then tells us to select the value of c (lake or grass) that makes $f_c(\vec{x})$ larger.

First we start by calculating the values of both discriminants:

$$\begin{aligned} f_{\text{lake}}\left(\begin{bmatrix}\text{overcast}\\ \text{day}\\ \text{full}\end{bmatrix}\right) &= \widehat{\Pr}(y = \text{lake}) \cdot \widehat{\Pr}(x_1 = \text{overcast} \mid y = \text{lake}) \\ &\qquad \cdot \widehat{\Pr}(x_2 = \text{day} \mid y = \text{lake}) \cdot \widehat{\Pr}(x_3 = \text{full} \mid y = \text{lake}) \\ &= 0.6 \cdot 0.17 \cdot 0.33 \cdot 0.5 \\ &= 0.02 \end{aligned} \tag{3.10}$$

$$\begin{aligned} f_{\text{grass}}\left(\begin{bmatrix}\text{overcast}\\ \text{day}\\ \text{full}\end{bmatrix}\right) &= \widehat{\Pr}(y = \text{grass}) \cdot \widehat{\Pr}(x_1 = \text{overcast} \mid y = \text{grass}) \\ &\qquad \cdot \widehat{\Pr}(x_2 = \text{day} \mid y = \text{grass}) \cdot \widehat{\Pr}(x_3 = \text{full} \mid y = \text{grass}) \\ &= 0.4 \cdot 0.5 \cdot 0.5 \cdot 0.75 \\ &= 0.08 \end{aligned} \tag{3.11}$$

Equation 3.7 tells us to report the value of c which results in the larger discriminant. Therefore,

$$f_{\text{NB}}\left(\begin{bmatrix}\text{overcast}\\ \text{day}\\ \text{full}\end{bmatrix}\right) = \text{grass}\ . \qquad \text{because } 0.08 > 0.02 \tag{3.12}$$

The classifier would make a similar calculation for other input vectors $\vec{x}$. If there were more than two classes (two locations for Sibonelo), there would be more than two such discriminant calculations, and f would select the location associated with the largest discriminant.

Further Considerations

The calculations for the discriminant involve multiplying a series of probabilities, each less than 1. The result is a number that can quickly underflow a computer's floating-point precision, resulting in problems. To circumvent this problem, frequently we calculate its logarithm instead:

$$\begin{aligned}\ln f_c(\vec{x}) &= \ln\left(\widehat{\Pr}(y=c)\prod_{j=1}^{n}\widehat{\Pr}(x_j \mid y=c)\right)\\ &= \ln\widehat{\Pr}(y=c) + \ln\prod_{j=1}^{n}\widehat{\Pr}(x_j \mid y=c)\\ &= \ln\widehat{\Pr}(y=c) + \sum_{j=1}^{n}\ln\widehat{\Pr}(x_j \mid y=c)\end{aligned} \tag{3.13}$$

Thus, instead of multiplying the probabilities together, we add their logarithms (in any base). Because ln is a monotonic function, it does not change which c corresponds to the maximum of the discriminant and therefore using $\ln f_c(\vec{x})$ instead of $f_c(\vec{x})$ in Equation 3.7 does not change the answer.

This transformation also reveals that naive Bayes can be seen as a scoring method[6] where we score each possible label by starting with its log-prior probability, $\ln\widehat{\Pr}(c)$, and adding a score for the value of each feature, $\ln\widehat{\Pr}(x_j \mid c)$. Whichever label scores highest is the "winner." This shows the effect of the naive Bayes assumption: Each feature is considered only in isolation (with the potential label, but not considering any other feature).

[6] like all of those magazine or online quizzes where you get points for each type of hippopotamus you have seen and at the end you compare your score to those of your friends to see who is the "real safari expert"

Naive Bayes, as presented here, assumes categorical features. If there are non-categorical features (such as counts, speeds, temperatures, or latitudes), the tabular forms of the empirical distributions must be generalized to handle them. We do not cover it here, but see Exercise 3.9 as an example of how to handle continuous features. Regardless of the feature types, naive Bayes is a classification method and not suitable for regression.

Exercises

Exercise 3.1

One percent of all hippopotamuses have a disease called rivermuddle syndrome which can cause them to be clumsy and dizzy. Fortunately, researchers have developed a test for this symdrome: they time how fast a hippopotamus can navigate an underwater obstacle course. This test has a 3% false-positive rate. That is, in 3% of hippopotamuses who do not have the syndrome, the test reports they do have rivermuddle syndrome. The test has a 2% false-negative rate. That is, in 2% of hippopotamuses with rivermuddle syndrome, the test reports they do not have the disease.

Let D be a random variable that represents whether a hippopotamus has rivermuddle syndrome. Let T be a random variable that represents whether the same hippopotamus tests positive for rivermuddle syndrome.

a. What is $\Pr(D = +)$?
b. What is the table for $\Pr(T \mid D)$?
c. What is the table for $\Pr(T, D)$?
d. What is the probability a hippopotamus that tests positive for rivermuddle syndrome has rivermuddle syndrome?

Exercise 3.2

Below is the joint distribution of hippopotamus physical characteristics.[7] We consider hippopotamuses who with mass more than 1,500 kg to be "heavy" and those less than 1,500 kg to be "light." We consider hippopotamuses taller than 1.5 m in height to be "tall" and those less than 1.5 m to be "short."

[7] While this is presented as the truth and you should take it as such for this question, these numbers are made-up. Apologies to all hippopotamuses.

mass (x_1)	height (x_2)	sex (y)	$\Pr(x_1, x_2, y)$
light	short	female	0.25
heavy	short	female	0.10
light	tall	female	0.20
heavy	tall	female	0.10
light	short	male	0.10
heavy	short	male	0.05
light	tall	male	0.05
heavy	tall	male	0.15

a. From the distribution above, what are the marginal distributions over each of the three random variables (x_1, x_2, y)?
b. What is the joint marginal distribution over x_1 and y?
c. What is the conditional distribution $\Pr(x_1 \mid y)$?
d. What is the conditional distribution $\Pr(y \mid x_1)$?
e. What is the Bayes-optimal rule for y based on x_1 and x_2?

Exercise 3.3

Consider predicting whether a student will pass an exam. You have two features: how much they studied and how much they slept the night before.

a. Why are these two features not independent?
b. Construct a reasonable joint distribution over all three variables (assume each are binary) for this scenario in which the two features are not independent, but are independent conditioned on the target (whether the student passes the exam).

Exercise 3.4

Consider a binary classification problem with n features, each with 3 values. For any distribution, we will define the number of parameters as the number of values that need to be estimated to fully specify the distribution. We will define the number of effective parameters to be the number of parameters minus the number of independent constraints.

a. How many parameters are there in the full conditional distribution $\widehat{\Pr}(\vec{x} \mid y)$? How many effective parameters are there?
b. If we instead use a naive Bayes model, how many parameters are there? How many effective parameters?

Exercise 3.5

You have been studying the elusive wumpus population. You have measured three things about the wumpuses you have found:

1. Age: young, adult
2. Eye color: red, green, brown
3. Size: short, medium, tall

You have also noted whether it is smelly or not. Your (training) dataset of ten posts is shown in Table 3.3.

a. Later, you run across a young, red-eyed, medium-sized, wumpus. Using naive Bayes, would you predict it would be smelly (yes) or not smelly (no)?
b. Even later, you peer through the dark and can only see the brown eyes of a wumpus? Is that enough to classify the wumpus as smelly or not smelly with naive Bayes? Why or why not?

age	eye	size	smelly?
young	red	short	yes
adult	red	medium	yes
young	red	medium	yes
adult	brown	short	yes
young	green	medium	no
young	brown	medium	no
young	red	tall	no
young	red	medium	no
adult	red	short	no
adult	brown	tall	no

Table 3.3: Wumpus dataset

Exercise 3.6

You have collected a (small) set of internet posts. You have measured three things about each:

1. Is it a repost: yes, no
2. Its length: short, long
3. Its view: postive (+), neutral (0), negative (-)

You have also classified each into one of three types: humor, review, and request. Your (training) dataset of twelve posts is shown in Table 3.4.

a. Based on the training data above, how would naive Bayes classify a post that is 1. not a repost, 2. long, and 3. of positive view (that is: no, long, +)?
b. Which of these features is the least useful for classifying the type of post using naive Bayes? Why?

repost?	length	view	type
no	short	+	humor
yes	long	0	humor
yes	short	-	humor
yes	long	-	review
no	short	+	review
yes	long	+	review
no	long	0	request
yes	long	-	request
no	long	-	request
yes	short	+	request
yes	short	+	request
yes	short	0	request

Table 3.4: Internet posts dataset

Exercise 3.7

You have surveyed the local jack-o-lantern population. You have measured three things about each:

1. Eye shape: ●, ▲, ■, ◆
2. Nose length: none, short, long
3. Mouth shape: toothy, smile, frown

You have also classified each into one of three types: funny, scary, and weird. Your (training) dataset of twelve jack-o-lanterns is as in Table 3.5.

Based on the training data above, how would naive Bayes classify a (testing) jack-o-lantern that has circle eyes, a long nose, and a toothy mouth (●, long, toothy)?

eyes	nose	mouth	type
◆	long	smile	funny
●	short	toothy	funny
■	long	smile	scary
●	short	toothy	scary
■	short	toothy	scary
▲	none	toothy	scary
■	none	frown	weird
◆	long	toothy	weird
▲	long	frown	weird
●	short	smile	weird
●	none	toothy	weird
●	short	toothy	weird

Table 3.5: Jack-o-lantern dataset

Exercise 3.8

Consider the case of naive Bayes for binary classification (where there are only two classes). Further, consider the discriminant in "log form," as in Equation 3.13.

If the two classes are "A" and "B," we have two log-discriminants, $\ln f_{\text{A}}(\vec{x})$ and $\ln f_{\text{B}}(\vec{x})$. Asking which one is larger is the same as asking whether their difference is greater or less than zero. Call this difference $\tilde{f}(\vec{x}) = \ln f_{\text{A}}(\vec{x}) - \ln f_{\text{B}}(\vec{x})$.

a. With this in mind, $\tilde{f}(\vec{x})$ can be written as a sum as follows:

$$\tilde{f}(\vec{x}) = \beta_0 + \sum_{i=1}^{n} \beta_i(x_i)$$

Write down the expressions for $\beta_0, \ldots, \beta_n$

b. If we compute $\tilde{f}(\vec{x})$ one term at a time (starting with β_0 and then adding each next term in turn: $\beta_1, \beta_2, \ldots, \beta_n$), under what condition can we stop the calculation early (and always still produce the same prediction as if we had fully calculated $\tilde{f}(\vec{x})$)? Be mathematically precise in writing down this condition.
c. Given the answer to part b, how could the features be reordered to allow for the best possible chance of early stopping?

Exercise 3.9

We can extend the naive Bayes method to allow for continuous (real-valued) features. If x_i is continuous, we then need to use $\hat{p}(x_i \mid y)$ (a conditional probability *density*) instead of $\widehat{\Pr}(x_i \mid y)$ (a conditional probability). When x_i was categorical (discrete-valued), we could represent $\widehat{\Pr}(x_i \mid y)$ as a table. For the density $\hat{p}(x_i \mid y)$ that is not possible because there is an infinite number of values of x_i.

One common solution is to model $\hat{p}(x_i \mid y)$ as a Gaussian distribution. It has a mean, $\mu_{i,y}$, and a standard deviation, $\sigma_{i,y}$, which are each particular to this feature (i) and this class label (y). They can be estimated as the empirical mean and standard deviation for feature i when the class is y.

Using the dataset in Table 3.6 (Sibonelo is trying to determine the type of a snack),

a. What are the estimated means and standard deviations for x_2 (roundness) and x_3 (juiciness)?
b. Using naive Bayes, how would Sibonelo classify a snack that was from a crate, had a roundness of 0.3, and had a juiciness score of 0.5?

origin	roundness	juiciness	type
farm	0.8	0.6	watermelon
farm	0.3	0.6	watermelon
crate	0.2	0.8	watermelon
farm	0.5	0.4	watermelon
farm	0.5	0.5	watermelon
crate	0.5	0.3	pumpkin
farm	0.9	0.7	pumpkin
farm	0.9	0.1	pumpkin
crate	0.1	0.2	pumpkin

Table 3.6: Hippopotamus snack dataset